

Islamic University

Faculty of Engineering

Department of Computer Eng. Tech.

Year No. Forth



الجامعة الإسلامية

كلية الهندسة

قسم هندسة تقنيات الحاسوب

المرحلة الرابعة

Name of lab.: Information Theory and Coding

اسم المختبر: نظرية المعلومات والترميز

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Lab. 2

2026-2027

Description of Experiment / Practice**1. Experiment / Practice No: 2****2. Title of Experiment / Practice: Entropy Measuring for Multi-Messages****3. Objectives of Experiment / Practice:**

- Understand the concept of entropy in Information Theory.
- Study entropy as the average information of a source.
- Calculate entropy for a source with multiple messages.
- Study the effect of message probabilities on entropy.
- Use MATLAB to calculate entropy.
- Compare entropy for different probability distributions.

4. Number of Required Hours: (2 hours)**5. Expected Outcomes of Experiment / Practice**

After completing this experiment, students will be able to:

- Define entropy.
- Calculate the entropy of a source with multiple messages.
- Explain the relationship between message probabilities and entropy.
- Compare entropy for different probability distributions.
- Use MATLAB to calculate entropy.

6. Language: English**7. Short Explanation (theory)****What is Entropy?**

In Information Theory, entropy measures the average amount of information produced by an information source.

If a source produces several messages:

$$x_1, x_2, \dots, x_n$$

with probabilities:

$$P(x_1), P(x_2), \dots, P(x_n)$$

then the entropy of the source is:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

Where:

- $H(X)$ = Entropy of the source
- $P(x_i)$ = Probability of message x_i
- n = Number of possible messages
- The unit of entropy is bit/message

Basic Idea

Entropy represents the average information of all possible messages.

The self-information of each message is:

$$I(x_i) = - \log_2 P(x_i)$$

Therefore:

$$H(X) = \sum_{i=1}^n P(x_i) I(x_i)$$

Example 1

Suppose an information source produces four messages with equal probabilities:

$$P(X) = [0.25, 0.25, 0.25, 0.25]$$

The entropy is:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

$$H(X) = -[4(0.25 \log_2(0.25))]$$

$$\text{Since: } \log_2(0.25) = -2$$

$$\text{Then: } H(X) = 2 \text{ bits/message}$$

Important Observation

When all messages have equal probabilities, the uncertainty is high and the entropy reaches its maximum value:

$$H_{max} = \log_2(n)$$

For four messages:

$$H_{max} = \log_2 4$$

$$H_{max} = 2\text{bits/message}$$

Example 2

Suppose another source has:

$$P(X) = [0.5, 0.25, 0.125, 0.125]$$

Then:

$$\begin{aligned} H(X) &= -[(0.5 \log_2(0.5)) + (0.25 \log_2(0.25)) + (0.125 \log_2(0.125)) \\ &\quad + (0.125 \log_2(0.125))] \\ &= -[0.5(-1) + 0.25(-2) + 0.125(-3) + 0.125(-3)] \\ &= 1.75\text{bits/message} \end{aligned}$$

8. Required Materials

- MATLAB Software
- Computer/Laptop
- Lecture Notes

9. Apparatus and Instruments

- Personal Computer
- MATLAB Installed Environment

10. Procedures

1. Start MATLAB.
2. Clear the command window, workspace variables, and previously opened figures.
3. Define the probabilities of the messages.
4. Verify that the sum of probabilities is equal to 1.
5. Calculate the self-information of each message using:

$$I(x_i) = -\log_2 P(x_i)$$

6. Calculate the entropy using:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

7. Display the probabilities and self-information values.
8. Display the entropy of the source.
9. Plot the probability and self-information of each message.
10. Observe the relationship between message probability and information.

MATLAB Code:

```
clc; clear; close all;  
P = [0.5 0.25 0.125 0.125];  
I = -log2(P);  
H = sum(P .* I);  
disp('Probability   Self-Information (bits)')  
disp([P' I'])  
disp('Entropy (bits/message)')  
disp(H)  
figure  
bar(I)  
grid on  
xlabel('Message')  
ylabel('Self-Information (bits)')  
title('Self-Information of Messages')
```

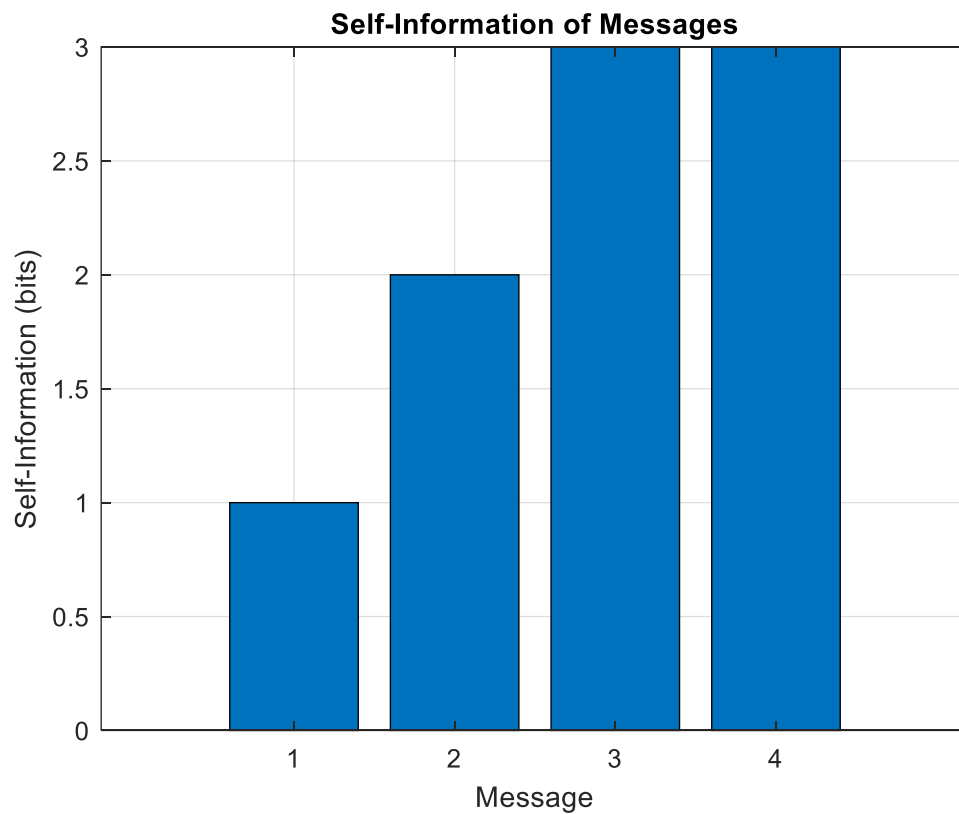
11. Observation / Result

The calculated values are:

Command Window

Probability	Self-Information (bits)
0.5000	1.0000
0.2500	2.0000
0.1250	3.0000
0.1250	3.0000

Entropy (bits/message)
1.7500



The calculated entropy is:

$$H(X)=1.75 \text{ bits/message}$$

The results show that messages with lower probabilities have higher self-information.

The entropy represents the average information produced by the source.

For equal message probabilities, the entropy reaches its maximum value.

12.Conclusion

In this experiment, the entropy of a source containing multiple messages was calculated using MATLAB.

The entropy depends on the probabilities of the messages and represents the average amount of information produced by the source.

Entropy is calculated using:

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i)$$

When all messages have equal probabilities, the entropy is maximum.

13.Homework

Q1: Using MATLAB, consider the following two information sources:

Source A = [0.25, 0.25, 0.25, 0.25]

Source B = [0.7, 0.1, 0.1, 0.1]

Calculate the entropy of each source and determine which source has higher entropy. Explain the reason based on the probability distribution.

14.References

- Claude E. Shannon, A Mathematical Theory of Communication.
- Thomas M. Cover and Joy A. Thomas, Elements of Information Theory.
- MATLAB Documentation.